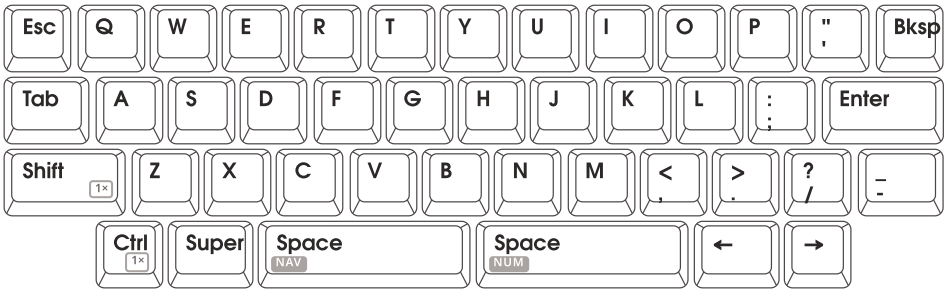
Agar mini

Custom ZMK Layout
Key Function Manual

ZMK FIRMWARE

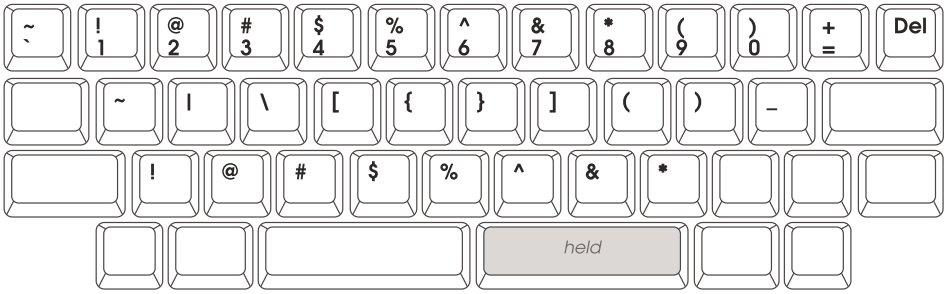
Quick tip: a **grey tag** is what the key does when held. **1×** marks a sticky key: tap it, then the next key. Blank keys pass through to Layer 0.

Layer 0 BASE



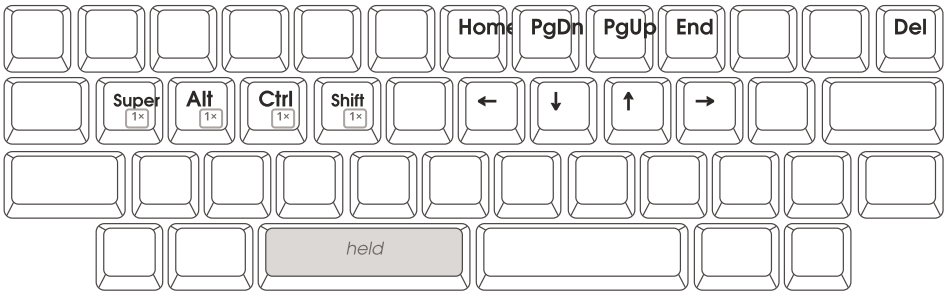
- 1× **Sticky:** tap, then the next key
- ' **Accent key:** tap it, then the vowel (Layer 4), or Space for '; Shift gives "
- NUM **Hold for Layer 1**, right Space
- NAV **Hold for Layer 2**, left Space
- **Both Spaces** for Layer 3
- Super **Plain key**, hold it
- Shift **The only Shift**, bottom-left as usual
- ← → **Arrows:** with Super, switch window; → accepts fish's suggestion

Layer 1 NUM



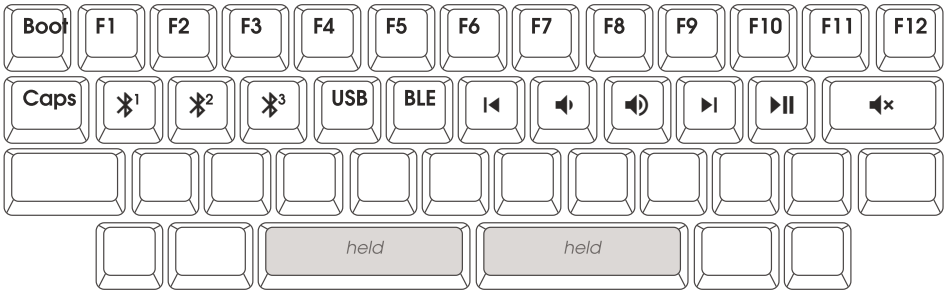
- **Hold right Space**
- 1–0 **Super + number** switches mango workspace
- . / - **Pass through** for ~/, 1.5, IPs
- ` ~ ^ **Plain characters**, not dead keys

Layer 2 NAV



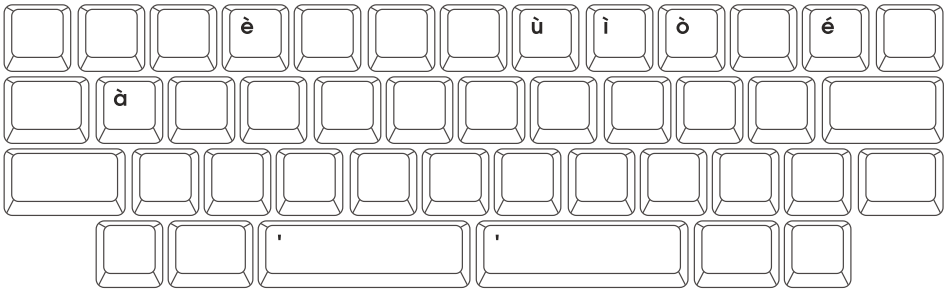
- **Hold left Space**
- ← ↓ ↑ → **Arrows on hjkl**
- 1× **Sticky modifiers:** tap D, tap L = Ctrl+→
- Home **Home, PgDn, PgUp, End** sit above the arrows

Layer 3 FN



- **Hold both Spaces**
- BT1 **Bluetooth profile 1–3**; hold 3 s to clear its pairing
- USB **Switch output** to USB or BLE
- Boot **Bootloader:** hold 3 s; a tap restarts
- ⏮ ⏯ ⏭ **Media** previous, play/pause, next
- 🔉 🔊 🔇 **Volume** down, up, mute

Layer 4 ACC



- 1× **Tap '**, then the key; expires after 1.5 s
- ' **Either Space** types the apostrophe; ' twice types é
- È **Capital:** tap Shift after '
- . **Other keys pass through**; the apostrophe is lost

Everyday actions

ACTION	SENDS	KEYS
è à ì ò ù		tap ', tap the vowel
é		tap ' twice
È		tap ', tap Shift, tap E
Backtick	`` ` ``	hold R Space + Esc
Capital letter		tap Shift (bottom left), tap the letter
Apostrophe / quote	' / "	tap ', tap Space / tap Shift, tap '
Newline in Claude Code	Shift+Enter	tap Shift, tap Enter
Question / exclamation mark	? / !	tap Shift, tap / / hold R Space + Z
Number	1 … 0	hold R Space + Q ... P
Backspace		top-right corner, as on the Q4
Run the last command again	Up, Enter	hold L Space + K, tap Enter
Copy / paste	Ctrl+C / Ctrl+V	tap Ctrl, tap C / V
Kitty copy / paste	Ctrl+Shift+C / V	tap Ctrl, tap Shift, tap C / V
Switch workspace	Super+3	hold Super + hold R Space + E
Move window to workspace	Super+Shift+3	tap Shift, then hold Super + hold R Space + E
Other mango shortcuts	Super+Return, Super+q	hold Super + the key
Focus left / right	Super+Left / Right	hold Super + ← / →
Focus up / down	Super+Up	hold L Space, tap A, tap K
Kitty tab	Alt+3	hold L Space, tap S (Alt), then hold R Space + E
Accept fish autosuggestion	Right	tap → (bottom right)

No mango config changes are needed: every action sends exactly what a full-size keyboard would.

Timing

KEY	FLAVOR	TERM	QUICK-TAP	PRIOR IDLE
Space / NAV	balanced	200	175	150
Space / NUM	balanced	200	175	—

STICKY KEYS	RELEASE AFTER	QUICK RELEASE
Shift, Ctrl, Alt, Super	1500	yes
' (sticky layer)	1500	no

- All values in milliseconds. Letters have no hold behaviour: they type on press.
- Sticky keys apply to the next key only and expire after 1.5 s. Tapping Ctrl then Shift stacks them. They can also be held normally. ' ignores modifiers, so Shift tapped between ' and the vowel still gives the capital; Shift before ' gives ". A key other than a vowel, ' or Space drops the apostrophe.
- Space / NUM has no prior-idle setting; otherwise typing `port 8080` quickly would give letters instead of digits.

Design

1. **Thumbs hold, nails tap one key at a time.** No chords; ring fingers only reach Esc and Backspace, the pinky only Shift, with the back of the finger.
2. **' is a dead key**, as on `us intl`: ' + vowel = è, ' + Space = ', with Italian grave accents.
3. **Sticky Shift, Ctrl and Alt:** tap, then the next key.
4. **" ` ~ ^ are plain**, despite `us intl` dead keys.
5. **Shift where it always is,** bottom-left. Super under the left thumb, ← → under the right. Backspace stays top-right.

Requires `us intl`

Accents and plain quotes are typed as `us intl` dead-key sequences: ` + vowel for è, ' + e for é, ' + Space for '. On a device with a plain US layout, accents come out as `e and quotes get an extra space.

Dropped from stock

Mouse keys (trackpad available), Win+V, Print Screen (mango uses Super+Shift+S), Insert, Pause, Scroll Lock, right Alt, arrow cluster (← → kept bottom right, all four on NAV).